From Agents to Sustainability: Charting the Evolution and Impact of Multi-Agent Systems in Circular Supply Chain Management

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Abstract: **The integration of Multi-Agent Systems (MAS) into supply chain management presents transformative opportunities for achieving circular economy (CE) goals and sustainability objectives. This bibliometric study systematically analyzes 281 peer-reviewed articles extracted from Scopus, Web of Science, and ScienceDirect databases (2005–2025), leveraging VOSviewer software for comprehensive analysis. Six thematic clusters emerged, revealing significant MAS applications across diverse domains: sustainable circular supply chains, urban waste management, closed-loop decision-making frameworks, industrial environmental management, specialized waste recycling optimization, and agricultural bioeconomy initiatives. The findings underscore MAS's vital role in promoting decentralized decision-making, resource optimization, lifecycle sustainability, and environmental resilience. Despite demonstrated benefits, challenges in scalability, real-time data integration, and socio-economic complexities highlight avenues for future research. This study provides valuable insights for researchers, industry practitioners, and policymakers, guiding further MAS-driven innovations for sustainable circular economy transitions.**

Keywords— Multi-Agent Systems, Circular Economy, Sustainable Supply Chain, Agent-based Modeling, Waste Management,

I. Introduction

Recently, the incorporation of Multi-Agent Systems (MAS) into supply chain management has garnered considerable interest owing to their decentralized characteristics and ability for autonomous decision-making, hence tackling the difficulties inherent in contemporary supply chains [1][2]. MAS are ideal for promoting circular economy (CE) initiatives including recycling, remanufacturing, and waste reduction since they have demonstrated significant promise in improving sustainability, efficiency, and adaptability [3][4].

By facilitating dynamic coordination, agent-based simulation, and distributed optimization over intricate networks, MAS provide a natural fit with the circular economy's emphasis on closed-loop flows, multi-stakeholder collaboration, and decentralized resource management. MAS is especially helpful in operationalizing circular strategies like reverse logistics, shared resource platforms, or urban mining because of its capacity to represent heterogeneous actors (such as suppliers, consumers, and regulators), mimic behavioral responses, and promote cooperation or negotiation.

Furthermore, the implementation of MAS is congruent with Industry 4.0 concepts, facilitating real-time data analytics, adaptive modeling, and intelligent coordination inside linked supply chain ecosystems [5][6].

The literature is still scattered, with relatively little systematic knowledge of the specific function of MAS in supporting circular economy projects across many sectors and operational environments, despite increased interest and encouraging study results [7][8]. Previous reviews have examined MAS from generic supply chain perspectives or general sustainability applications, yet far-reaching bibliometric observations explicitly linking MAS application to circular economy configurations are few [9]. Consequently, there is a critical knowledge gap concerning the detailed thematic structure, key trends, and evolving patterns in MAS-driven circular economy research.

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To bridge this literature gap and provide structured insights, the present bibliometric analysis systematically reviews the scholarly literature from 2005 to 2025, aiming to clarify the following critical research questions:

Q1: What are the emerging thematic areas and trends characterizing the integration of MAS within circular economy and sustainable supply chains?

Q2: Which regions, countries, and scholarly sources have most significantly influenced and shaped the intellectual structure of MAS-driven circular economy research?

Q3: What key gaps and challenges remain unaddressed in current MAS applications to the circular economy and sustainability within supply chain management, and how can future research address these challenges?

By answering these basic questions, this research does not only synthesize current scholarly offerings but also discerns future directions, offering valuable perspective to researchers, practitioners, and policymakers involved in sustainable and circular supply chain management.

II. Literature Review

Using multi-agent systems (MAS) in the supply chain management context has attracted much academic interest, especially in the context of circular economy (CE) and sustainable operations. MAS offer a decentralized and intelligent approach to managing the complexity of contemporary supply chains through efficient resource utilization, waste reduction, and sustainable operations. [10][11]. Recent studies underline the importance of MAS to support collective decision-making and to provide operational flexibility in sustainable supply chain networks [12][13].

In the sense of circular economy, MAS support strong reverse logistics system and close loop supply chain process, which greatly help recycling, remanufacture and waste management. For this purpose, agent-based modeling and simulation have become indispensable, allowing organizations to experiment and confirm sustainable operations strategies prior to their actual application [14]. Furthermore, MAS-based applications in urban waste management also successfully manage the challenges in e-waste recycling, garbage disposal as well as MSWM by providing cities and governments with potential instruments to policy development and sustainable transitions [15].

In industrial and construction applications, the MAS-based simulation methodologies have improved lifecycle analysis, energy and emissions control and management, and have played a key role in environmental and sustainable industrial practices [16]. However, the practical challenges still remain, including scalability, real-time data integration, and computational complexity, indicating the need for future studies on potential adaptive MAS architectures for large-scale, realistic industrial environments [17].

In addition, customized MAS have proven to be effective in managing intricate situations faced in nuclear waste disposal, waste recovery and prevention of environmentally risky malpractices [18]. While enjoying these benefits, the complexity of regulations, economy, safety are the reasons that should require economical MAS techniques that comply with strict specific industry standards [19].

MAS has also been applied in the agro-sector and in the bioeconomy sector, with good outcomes, in terms of more efficient material use, water recycling, and sustainable agricultural means [20]. Nonetheless, lop-sided technological capabilities and resource barriers represent significant issues which call for tailored MAS solutions to address regional level differences also with respect to sustainable development [21].

The literature highlights MAS's transformative potential for circular economy and sustainability in a variety of domains, such as urban management, industry, construction, agriculture, and specialized high-risk domains. Despite the strides made, future research needs to address remaining practical challenges, particularly relating to scalability, integration complexity, regulatory and legislation, and socio-economic issues, in providing MAS solutions, ensuring MAS solutions can effectively support global sustainability transitions.

III. Methodology

To construct the bibliometric dataset, this study conducted a systematic search across three major academic databases: Scopus, Web of Science (WoS), and ScienceDirect, targeting peer-reviewed journal articles published in English between 2005 and March 2025. The search strategy specifically identified literature at the intersection of Multi-Agent Systems (MAS), Circular Economy (CE), and sustainable supply chain management, employing tailored keyword combinations for each database. To ensure dataset quality and relevance, only peer-reviewed journal articles directly addressing MAS applications within CE and supply chain sustainability contexts were included, while conference papers, book chapters, and non-peer-reviewed materials were excluded. Duplicate records were carefully removed, resulting in a final dataset of 281 articles, analyzed using VOSviewer software for bibliometric mapping.

However, this approach potentially introduces certain biases, particularly regarding comprehensiveness. By excluding non-English and non-peer-reviewed sources, valuable insights published in other languages or formats may have been overlooked. Consequently, this might impact the generalizability of findings, particularly concerning regions or contexts underrepresented in English-language scholarship.

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Additionally, the publication data for 2025 may be incomplete due to the search cut-off in March 2025.

To ensure transparency and reproducibility, the keyword strategy was carefully constructed to target studies explicitly addressing the integration of MAS in circular or sustainable supply chain contexts. Although the full search strings are not reproduced here due to space constraints, they systematically combined terms related to MAS (e.g., "multi-agent system", "agent-based model"), circular economy (e.g., "reverse logistics", "resource recovery"), and supply chain management (e.g., "value chain", "logistics"). This refined selection ensured thematic consistency across the dataset and alignment with the study's focus on decentralized intelligence applied to circular economy systems.

IV. Results and Discussion

A. *Temporal Trends in MAS-Driven Circular Supply Chain Research: Publication Analysis by Year*

The publication trends illustrated in Figure 1 highlight a steady and progressive rise in scholarly interest regarding the integration of Multi-Agent Systems (MAS) in the context of circular and sustainable supply chains. From 2005 to 2014, academic output remained low, with fewer than five documents published annually. This initial phase reflects the early stage of MAS development and the limited adoption of circular economy principles in supply chain research.

A notable shift occurred beginning in 2015, where the number of publications started to increase gradually, with modest but consistent growth through 2020. This growth phase aligns with the expanding awareness of sustainability challenges and the emergence of intelligent systems for decentralized supply chain coordination. During this period, publications rose from 8 in 2015 to 19 in 2020, setting the stage for more focused research.

From 2021 to 2024, the industry experienced a steep growth in academic production, a span of exponential growth. Publications increased rather dramatically from 33 in 2021 to a high of 60 in 2024. This increase represents a broader push for digitalization, the acceleration of Industry 4.0 practices, and growing interest in intelligent, agent-based techniques for coordinating complicated, closed-loop supply chains. It also shows global policy direction towards sustainable transitions to circular economics and development.

In 2025, the number of publications see a temporary decline to 25. However, this seeming drop is not a sign of diminishing interest but instead a function of the data cutoff date which took place in March 2025. As it stands, this year's publication record is poor and expected to increase by the end of the year. As such, the current year's publication count is incomplete and expected to rise by the end of the year.

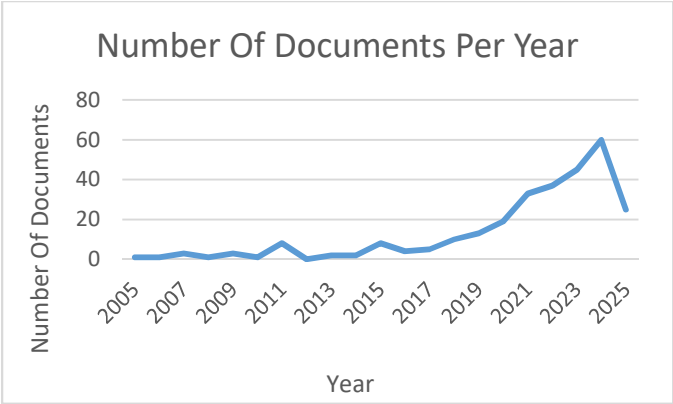

Fig. 1. Annual Publication Trends,

B. *Geographical Contributions to MAS-Driven Circular Economy and Supply Chain Research: A Country Analysis*

As depicted in Figure 2, the geographical analysis of contributions reveals a globally distributed yet concentrated research effort around Multi-Agent Systems (MAS) in the context of circular economy and supply chain management. China emerges as the leading contributor with 42 publications and 763 citations, positioning itself as a key driver of research in this niche domain. Its dominance is further reinforced by its high total link strength (13), signifying strong international research collaborations, particularly with countries such as the United Kingdom, Canada, and the Netherlands.

The United Kingdom and the United States follow with 16 and 15 publications respectively, each exceeding 250 citations, underlining their sustained academic leadership in agent-based systems and sustainable operations. Notably, the Netherlands and Italy, despite a slightly lower publication volume (13 each), show remarkably high link strengths (17 each), suggesting dense collaboration networks and a substantial impact on the intellectual structure of the field.

France, Germany, and Canada also contribute significantly, combining both respectable publication outputs and moderate citation scores, indicative of their focus on interdisciplinary approaches and circular logistics frameworks. Other European countries including Spain, Sweden, and Switzerland contribute with smaller volumes, yet maintain active collaboration pathways, as reflected in their visualization links.

Emerging economies such as India and Iran demonstrate increasing engagement, with India registering 6 publications and 243 citations, highlighting its evolving presence in MAS applications for sustainable development. Morocco appears with 4 publications, reflecting a nascent but growing academic interest in circular economy optimization from the MENA region.

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The network visualization underscores three prominent collaboration clusters: China-centric partnerships (blue cluster), a strong European bloc (green-yellow), and a smaller, emerging collaboration space linking countries like Denmark, Sweden, and India (red cluster). The Netherlands, in particular, acts as a vital collaboration hub, connecting Eastern and Western research efforts with the highest total link strength among all countries.

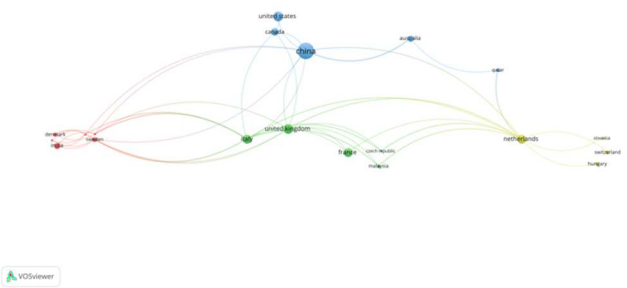

Fig. 2. Country co-citation map of MAS and circular economy research,

This cross-national engagement illustrates the increasing importance of MAS in addressing sustainability and efficiency in global supply chains. The network suggests a shift from isolated national studies toward more integrated and collaborative approaches, responding to the complexity of circular economy transitions in logistics and production systems.

C. *Scholarly Impact and Contributions: A Journal Analysis of MAS-Driven Circular Economy and Supply Chain Research*

As illustrated in Table I and Figure 3, the analysis of the most influential journals contributing to MAS-driven research in the context of circular economy and supply chain management reveals a clear interdisciplinary landscape. The leading source is the *Journal of Cleaner Production*, with 20 publications and a remarkable 688 citations, demonstrating its pivotal role in fostering sustainability-oriented research and innovative practices for circular supply chains.

TABLE I. TOP 5 CONTRIBUTING JOURNALS IN MAS-DRIVEN CIRCULAR ECONOMY AND SUPPLY CHAIN RESEARCH

Rank	Journal Title	Documents	Citations
1	*Journal of Cleaner Production*	20	688
2	*Resources, Conservation and Recycling*	5	303
3	*ACS Sustainable Chemistry and Engineering*	3	195
4	*Applied Clay Science*	2	155
5	*Sustainable Production and Consumption*	2	101

Resources, Conservation and Recycling ranks second, contributing 5 publications and receiving 303 citations. This reflects its strategic focus on sustainable resource management, circular economy practices, and environmental efficiency, core aspects of MAS applications in modern supply chains.

ACS Sustainable Chemistry and Engineering and *Applied Clay Science* follow with 3 and 2 documents respectively, both highlighting the technological and material science dimensions associated with sustainability-driven supply chains. Notably, the *Sustainable Production and Consumption* journal, with 2 documents and 101 citations, underlines the increasing scholarly attention given to consumption patterns and their interaction with sustainable production systems, areas where MAS plays a coordination role.

These findings reflect a growing scholarly interest in integrating MAS into circular supply chains, spanning a diverse set of scientific disciplines, from environmental management to production engineering and information systems.

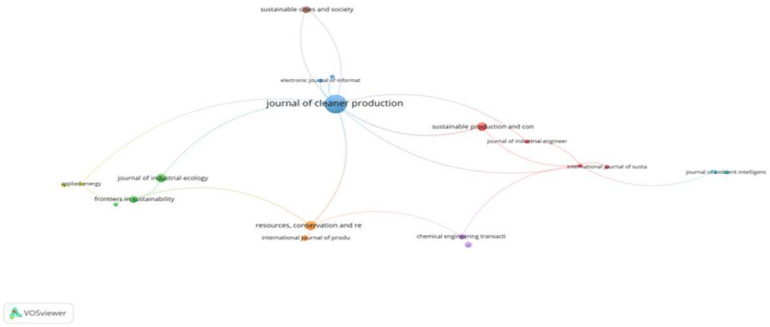

Fig. 3. Co-Citation Network of Core Journals in Multi-Agent Systems and Circular Economy Research.

D. *Key Themes and Trends in MAS-Driven Circular Economy and Supply Chain Research: A Keyword Analysis*

The keyword analysis demonstrates major themes and patterns bringing together
Multi-Agent Systems (MAS), circular economy, and supply chain management. "Circular economy" (42 occurrences) overcomes, highlighting the central role in closed-loop, efficient systems. Important high-frequency keywords like "waste" management' (38), 'autonomous agents' (35), 'recycling (34), and "sustainable development" (32) underscore MAS's contribution towards decentralized decision-making and performance. The focus on "agent-based" model" (31), "computational methods" (31), and "simulation" (21) points to the methodological reliance on simulation to manage system complexity. New digital trends like "Industry 4.0" and "system dynamics" indicate a shifting research focus. Overall, MAS emerges as a transformative

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an enabler of sustainability and intelligence in modern supply chains.

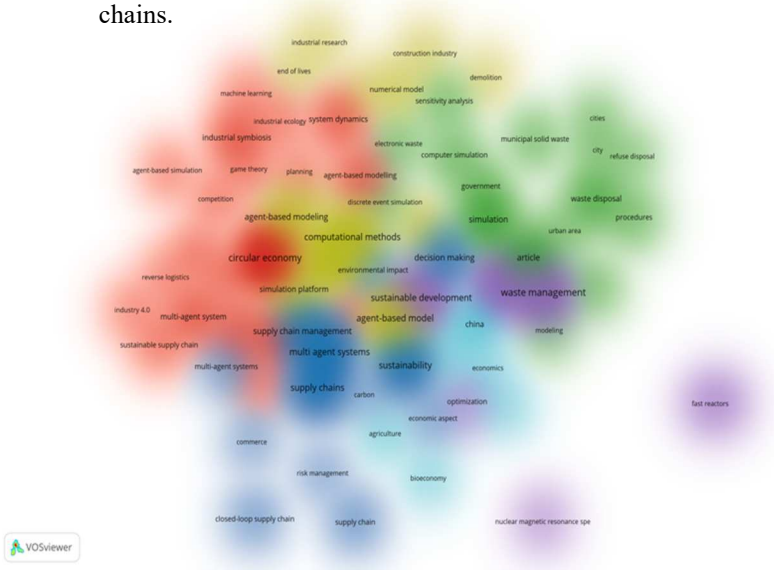

Fig. 4. Keyword Co-Occurrence Network Visualizing Research Themes.

E. Thematic Clustering of MAS-Driven Circular Economy and Supply Chain Research: Key Areas and Insights

The cluster analysis clearly reveals that Multi-Agent Systems (MAS) play a central role in enhancing circular performance, particularly through their ability to efficiently manage complex flows of products and materials. The intrinsic capacity of MAS to decentralize decision-making, coordinate interactions, and simulate dynamic scenarios makes this technology especially well-suited for operationalizing the principles of the circular economy. For instance, in the fields of reverse logistics and industrial recycling, MAS enable effective coordination among multiple stakeholders, thereby reducing operational costs and the environmental impacts associated with the reuse of materials and resources.

1) Cluster 1: Multi-Agent Systems for Sustainable Circular Supply Chains and Industrial Symbiosis:

Cluster 1 highlights the strategic integration of Multi-Agent Systems (MAS) into circular supply chains, where agent-based modelling, simulation, and decentralized coordination enable complex stakeholder interactions and sustainable operations. Industry 4.0 technologies further support MAS by enabling real-time data sharing and system interoperability [23], while concepts such as life cycle assessment, industrial symbiosis, and reverse logistics illustrate MAS's role in enhancing resource recovery and waste reduction [24].

MAS-based decision support systems have proven valuable in optimizing resource allocation and planning [25], and recent work emphasizes the potential of MAS–machine learning integration to improve forecasting and operational performance [26]. Economic dimensions such as cost analysis and game theory are increasingly used to manage sustainability–profitability trade-offs [27].

However, few studies have been conducted on practical models with limited real-world applications at scale. The complexity of hybrid MAS systems demand interdisciplinary solutions that unite technological, governance, and socio-economic insights for ensuring practical, scalable, and equitable implementations aligned with circular economy objectives.

For instance, MAS has been utilized to manage autonomous supply chains, enabling decentralized decision-making and adaptive coordination in complex manufacturing networks [22].

TABLE II: THEMATIC CLUSTERS OF KEYWORDS AND THEIR ASSOCIATED RESEARCH FOCUS AREAS

Cluster 1	Agent-based modelling, Agent-based simulation, Circular economy, Competition, Costs, Decision support system, Environmental economics, Game theory, Industrial ecology, Industrial symbiosis, Industry 4.0, Life cycle, Life cycle assessment, Machine learning, multi-agent system, Planning, Reverse logistics, Sustainable supply chain, Sustainable supply chain management, System dynamics	MAS and Circular Economy for Sustainable Supply Chain Management
Cluster 2	Cities, City, Computer simulation, Discrete event simulation, electronic waste, Government, Human, Modeling, Municipal solid waste, Procedures, Refuse disposal, Sensitivity analysis, Simulation, Urban area, Waste disposal	Simulation and Waste Management in Urban and Industrial Context
Cluster 3	Carbon, Closed-loop supply chain, Commerce, Decision making, Economic aspect, Environmental impact, multi-agent systems, Risk management, Supply chain, Supply chain management, Supply chains, Sustainability	MAS for Closed-Loop Supply Chain and Environmental Impact Management
Cluster 4	Agent-based model, Agent-based modeling, Autonomous agents, Computational methods, Construction industry, Demolition, End of lives, Energy utilization, Environmental management, Gas emissions, Industrial research, Numerical model, Simulation platform	MAS and Simulation Models for Environmental Management and Industrial Research
Cluster 5	Actinides, Fast reactors, Nuclear magnetic resonance, Optimization, Recycling, Sustainable development, Waste management	Optimization and Recycling in Sustainable Development
Cluster 6	Agriculture, Bioeconomy, China, Economic and social effects, Economics, Waste water recycling	Circular Economy and Bioeconomy for Sustainable Resource Management

2) Cluster 2: MAS Applications in Urban Waste Management and Simulation Modeling:

Cluster 2 centers on the application of Multi-Agent Systems (MAS) in urban and municipal waste management, where key

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terms like municipal solid waste, electronic waste, and refuse disposal reflect MAS's growing relevance in urban sustainability. MAS combined with discrete event and computer simulation has shown promise in optimizing waste collection and resource allocation, significantly reducing environmental impact and operational costs. An illustrative example is the MAS-driven discrete event simulation used to optimize urban rail transit passenger flow management, significantly improving efficiency and operational cost savings [28].

Agent-based models have been particularly effective in managing electronic waste recycling networks, enhancing regulatory compliance and traceability [29], while also providing valuable tools for city governments in policy evaluation and implementation [30]. Additionally, MAS applications in urban planning support behavioral modeling and coordination among diverse stakeholders [31].

However, despite these benefits, several critical limitations remain. Accurately simulating human-agent interactions continues to be a major challenge, as MAS models often oversimplify behavioral responses to waste policies, limiting their practical effectiveness. Moreover, the scalability of these systems from pilot projects to full-scale urban environments is hindered by integration issues with existing infrastructure and technologies. Addressing these gaps requires further research into adaptive and interoperable MAS architectures that better reflect real-world social dynamics and support scalable deployment in complex urban systems.

3) Cluster 3: MAS in Decision-Making for Sustainable and Closed-Loop Supply Chains:

Cluster 3 emphasizes the use of Multi-Agent Systems (MAS) in supporting sustainable and closed-loop supply chain decision-making. Keywords such as closed-loop supply chain, sustainability, and environmental impact highlight the integration of ecological and economic priorities into operational strategies. MAS frameworks have been shown to optimize resource use, improve efficiency, and reduce environmental footprints through decentralized coordination and intelligent decision support systems. A practical example is the MAS-based dynamic closed-loop supply chain framework, which integrates economic and ecological decisions, successfully applied in industrial scenarios to optimize sustainability outcomes [32].

The inclusion of terms like risk management and economic aspect reflects the importance of MAS in enhancing supply chain resilience and financial sustainability, particularly under uncertainty [33]. Moreover, MAS has been increasingly applied to manage carbon emissions, aligning operational choices with broader climate goals [34].

However, despite these advancements, critical challenges remain, particularly in terms of implementation complexity, real-time data integration, and scalability of MAS in large and dynamic supply chains [35]. Addressing these limitations is essential for translating MAS-based sustainability models into actionable, real-world solutions.

4) Cluster 4: Agent-based Modeling for Environmental Management in Industrial and Construction Contexts:

Cluster 4 highlights the application of agent-based modeling and computational methods in industrial environmental management, particularly within the construction sector and its lifecycle stages such as demolition and end-of-life processes. Keywords like energy utilization and gas emissions emphasize MAS's role in optimizing energy efficiency and minimizing environmental impact.

MAS facilitates the modeling of complicated building processes, enabling maximum use of resources and emissions reduction. Moreover, MAS-based numerical models offer critical assistance for strategic planning and sustainability transitions in industrial systems [36]. Such tools are increasingly vital to matching building practices to circular economy principles. Nonetheless, despite these advantages, notable challenges persist, especially regarding the reliance on real-time industrial data, which remains difficult to obtain and blend frequently between companies. For instance, MAS has been effectively deployed to plan and schedule precast component assembly in construction sites, reducing environmental impacts through optimized resource use and emissions management [37].

In addition, MAS computational complexity simulations can restrict their scalability in large-scale construction environments. Surmounting these challenges will be at the core of unlocking the complete environmental and operational advantages of MAS in industrial applications.

5) Cluster 5: MAS Applications in Waste Recycling and Optimization for Sustainable Development:

Cluster 5 emphasizes specialized applications of Multi-Agent Systems (MAS) in recycling, waste management, and optimization, all within the broader scope of sustainable development. The presence of technical terms like actinides, fast reactors, and nuclear magnetic resonance signals the deployment of MAS in complex and high-risk environments, particularly in nuclear waste management.

A prominent real-world example is the MAS integration in subway power supply systems for automated fault diagnosis, isolation, and recovery, improving reliability and sustainability in high-risk infrastructures [38]. This illustrates how MAS can enhance the efficiency of recycling and resource

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recovery operations in these sensitive contexts, contributing greatly to waste reduction and sustainability objectives. Despite this promising potential, substantial implementation hurdles still exist, most prominently the excessive cost, complexity, and strict regulations and rules pertaining to safety-critical domains like nuclear waste management. Future initiatives should be aimed at developing cost effective, largescale MAS architectures that include safety compliance and regulatory adaptability, enabling their practical application in hazardous industrial environments.

6) Cluster 6: MAS in Agriculture and Bioeconomy for Sustainable Economic and Social Impacts:

Cluster 6 highlights the role of Multi-Agent Systems (MAS) in advancing sustainability across agriculture, bioeconomy, and wastewater recycling. These applications reflect MAS's capacity to improve resource allocation, boost productivity, and support circular initiatives within agricultural systems. Studies demonstrate MAS-driven improvements in managing agricultural inputs and coordinating wastewater recycling, particularly within region-specific contexts such as China. Yet, despite encouraging outcomes, practical challenges persist, especially in rural settings where technological infrastructure is limited, and socioeconomic disparities can hinder adoption. A notable case study is the MAS-based management approach to the Water–Energy–Food Nexus in Ningdong, China, enhancing sustainable resource allocation and improving regional socio-economic outcomes [39]. To ensure equitable and scalable agricultural sustainability, future research should prioritize MAS implementations that accommodate local resource constraints and promote inclusive, community-sensitive bioeconomic development.

V. Conclusion

This bibliometric study examined the convergence of Multi-Agent Systems (MAS) for circular economy (CE) and sustainable supply chain management by analyzing 281 articles published from 2005 to 2025. Major findings identified six thematic clusters highlighting MAS functions in various domains, such as circular sustainable supply chains, urban waste management, closed-loop decision-making, industrial environmental techniques, specialized waste recycling, and agricultural bioeconomy processes. MAS exhibited considerable effectiveness in facilitating decentralized decision-making, optimizing resource allocation, enhancing waste reduction, and enabling lifecycle sustainability.

Practical implications of these findings provide industry stakeholders with strategic insights into effective MAS integration within circular supply chain management. Practitioners and policymakers can leverage these insights to enhance sustainability initiatives, reinforce regulatory compliance, streamline waste management operations, and foster innovative circular economy practices at organizational and regional levels.

However, several limitations should be acknowledged. First, bibliometric methods depend significantly on database coverage and keyword selection, which might omit pertinent studies. Second, this analysis included only peer-reviewed English-language journal articles, potentially excluding valuable insights from other languages and document formats, thus limiting generalizability.

Moreover, several technical implementation challenges remain critical barriers to the widespread adoption of MAS in practical settings. Scalability issues arise due to increasing computational complexity, particularly in large-scale MAS implementations where numerous agent interactions and real-time decision-making escalate computational demands and integration complexity. Furthermore, interoperability between MAS and legacy IT infrastructures remains challenging due to heterogeneous data standards, varying system architectures, and inconsistent communication protocols. To mitigate these issues, adopting standardized data exchange frameworks such as JSON or XML, coupled with middleware platforms (e.g., Apache Kafka or MQTT-based solutions), can significantly enhance real-time data integration, synchronization accuracy, and reduce data latency.

Future research is therefore recommended to specifically target these technical challenges, exploring innovative solutions for MAS scalability, interoperability standards, and effective real-time data integration methods. Additionally, systematic literature reviews and empirical studies focusing on real-world MAS deployments could further validate theoretical findings and provide deeper insights into practical sustainability and circular economy applications.

Finally, this study directly contributes to the scope of our doctoral research, particularly focusing on decentralized intelligence tools and their application within collaborative forecasting frameworks in circular economy contexts. As part of my ongoing thesis work, future research will aim to further investigate MAS-based models for enhancing collaboration, coordination, and sustainability performance in circular supply chains.

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